Question #1 of 20

Question ID: 1473606

It is *most accurate* to state that the upfront payment associated with a credit default swap (CDS) is:

A) sometimes made by the credit protection seller to the credit protection buyer.

B) always zero due to the way CDS are priced at origination.

C) greater when the reference obligation is high-yield debt rather than investment-grade debt.

Explanation

The CDS upfront payment may either be from the protection buyer to the seller, or vice-versa. If the credit spread is equal to the coupon rate, the upfront payment can be zero. CDS are valued by calculating the difference between the present value of the protection leg, versus the present value of the payment leg. The amount of upfront payment depends on the difference between the credit spread on the reference obligation and the CDS coupon rate, and hence need not be higher for a high-yield bond compared to an investment grade bond.

(Module 29.2, LOS 29.c)

Question #2 of 20

Question ID: 1473609

In anticipation of an announcement of leveraged buyout of a publicly traded company, which of the following actions would be *most appropriate*?

A) Buy the stock of the company and buy CDS protection on company's debt.

B) Sell protection of the company's bond and buy put options on the company's stock.

C) Buy both the stock and the bonds of the company.

Explanation

In the case of a leveraged buyout (LBO), the firm will issue a great amount of debt in order to repurchase all of the company's publicly traded equity. This additional debt will increase the CDS spread because default is now more likely. An investor who anticipates an LBO might purchase both the stock and CDS protection, both of which will increase in value when the LBO happens.

(Module 29.3, LOS 29.e)

Question #3 of 20

Question ID: 1473602

Regarding CDS credit events, a CDS is *least* likely to pay off upon occurrence of a:

A) bankruptcy

B) failure to pay

C) restructuring.

Explanation

CDS pay off upon occurrence of a credit event, which includes failure to pay, and bankruptcy. Restructuring is not considered a credit event in some countries (such as the United States, where bankruptcy is the preferred route.) Restructuring refers to events such as: reduction or deferral of principal or interest, change in the currency in which principal or interest will be paid, or change in an obligation's seniority or priority.

(Module 29.1, LOS 29.b)

Idrissa Sylla and Joel Lynch both work for Kazenga Asset Management. The fund made losses on fixed income securities during the 2008 credit crunch and is keen to minimize the risk of losses due to credit events going forward. Sylla and Lynch have been tasked with writing a report on the hedging of credit risk for the firm's investment committee. Extracts of their report are included below.

Introduction:

Credit Default Swaps (CDS) have the advantage of allowing the investor to separate credit risk and interest rate risk. Purchasing a CDS allows us to go long only the bond's credit risk.

A single-name CDS allows us to purchase credit protection for a single reference entity. Typically, the reference obligation for a single-name CDS is a senior unsecured bond. Interestingly there is a payoff not only when the reference obligation defaults but when any bond of the issuer that ranks pari passu with the reference obligation defaults.

The payoff received on a default will be the par value (notional principal) of the reference obligation less the value of the reference obligation after the credit event. Settlement after a credit event will either be physical delivery of the cheapest to deliver bond or alternatively a cash settlement.

Illustration CTD:

A credit event occurs for a single-name CDS with a three-year, senior bond as the reference obligation. The notional principal is $15m. Exhibit 1 shows bonds currently outstanding for the reference entity.

Exhibit 1: Current Market Price of Reference Entity Bonds

Bond type	Price
Bond Q: Subordinated unsecured 5-year maturity	30% of par
Bond P: Senior unsecured 2-year maturity	45% of par
Bond R: Senior unsecured 3-year maturity	50% of par

Value after Inception of CDS:

At initiation of a CDS, the CDS spread depends on the credit quality of the reference obligation at that point. Subsequent changes in credit quality of the reference obligation result in a gain or loss for the CDS holder. Entering an offsetting contract can monetize this gain (or loss). Exhibit 2 shows an illustration.

Exhibit 2: Illustration

Notional principal covered £36,000,000			
	Coupon rate	**Duration**	**Upfront Premium**
At initiation	1%	5	5%
1 year later	1%	4	8%

The Credit Curve:

The credit curve is the relationship between credit spreads and bond maturities for the same reference entity. Longer maturity bonds typically have a higher credit spread than shorter maturity bonds.

An investor purchases a 'naked' CDS when the investor does not hold the reference obligation. Essentially, it is a pure bet on the credit prospects of the bond issuer. If we believe the credit quality of the issuer will deteriorate and hence the credit curve steepens, we should take a short position in the CDS.

A curve trade is a type of long/short trade where the investor buys and sells protection on the same reference entity but with a different maturity. For example, if we were concerned about the credit risk in the short term but felt the entity's long term prospects were stronger, we would sell protection in a short maturity CDS and buy protection in a long maturity CDS. The improvement in the credit quality over time should cause the credit curve to flatten, resulting in a profit on the strategy.

Question #4 - 7 of 20

Question ID: 1473597

Which of the three statements in the introductory paragraph is correct?

A) The statement describing the separation of credit and interest rate risk.

B) The statement describing the bonds covered by a single name CDS.

C) The statement describing the payoff on a credit event.

Explanation

CDS do allow the separation of credit risk and interest risk on a bond. A long position in a CDS buys the protection against credit events and hence the investor is short credit risk.

Typically, a CDS will produce a pay off when any bond that ranks pari passu (same seniority) of the reference entity defaults.

The payoff on a credit event is the notional principal of the reference obligation less the market value of the cheapest to deliver bond. The cheapest to deliver bond must have the same seniority as the reference obligation.

(Module 29.1, LOS 29.a)

Question #5 - 7 of 20

Question ID: 1473598

Using the information under the heading "Illustration CTD," which of the three bonds would be the cheapest to deliver?

A) Bond Q.

B) Bond P.

C) Bond R.

Explanation

Bond Q is trading at the lowest price but the cheapest-to-deliver bond must rank pari passu with the reference obligation. Note that this is a CDS on a senior reference obligation and this bond is subordinated.

Bond R has the same seniority as the reference obligation but trades at a higher price than Bond P. Note that there is no requirement for the CTD bond to have the same maturity as the reference obligation.

Bond P has the same seniority as the reference obligation and trades at the lowest price. Payoff \$15m – (0.45)(\$15m) = \$8.25M.

(Module 29.1, LOS 29.a)

Question #6 - 7 of 20

Question ID: 1473599

Using information in Exhibit 2, the gain on the position is *closest* to:

A) £1,080,000.

B) £1,440,000.

C) £4,320,000.

Explanation

Step 1:

Calculate the CDS spreads:

upfront premium (%) = (credit spread – CDS coupon) × duration

credit spread = (upfront premium / duration) + CDS coupon

At initiation: credit spread = (5 / 5) + 1 = 2%

1-year later: credit spread = (8 / 4) + 1 = 3%

Step 2:

Compute the approximate profit to the buyer as:

profit for protection buyer ≈change in spread × duration

≈(0.03 – 0.02) × 4 × £36,000,000

profit ≈£1,440,000

(Module 29.1, LOS 29.a)

Question #7 - 7 of 20

Question ID: 1473600

Which of the comments relating to the credit curve is *least* accurate?

A) The definition of the credit curve.

B) The description and example of the naked CDS position.

C) The description and example of the curve trade.

Explanation

The definition of the credit curve is accurate.

The definition of the naked CDS position is also correct. Be careful here because the purchaser of the CDS is said to be taking a short position in credit risk. This seems counter intuitive as we normally describe a purchaser as long and a seller as short.

The definition of a curve trade is correct, however, the example is incorrect. If we believe that the credit condition of the reference entity will improve over time we should purchase protection in a short maturity CDS and sell protection in a long maturity CDS.

(Module 29.1, LOS 29.a)

Question #8 of 20

Question ID: 1473607

Suppose that an investment-grade bond's five-year credit spread is 175 bps, and the duration of the associated CDS is four years. Assuming a 1% CDS coupon, the upfront premium (expressed as a percentage of the notional) required to purchase five-year CDS protection on the company's debt will be *closest* to:

A) 8%

B) 3%

C) 4%

Explanation

To buy 5-year CDS protection, an investor would have to pay upfront the present value of the difference between the 100bps coupon and the current market spread of 175 bps. In this case, the upfront premium would be: Upfront premium ≈ (Credit spread – Fixed coupon) × Duration = (175bps – 100bps) × 4 = 3% of the notional.

(Module 29.2, LOS 29.c)

Question #9 of 20

Question ID: 1473590

Considering the two parties to a credit default swaps (CDS), the protection buyer is *most likely* to be:

A) bullish on the financial condition of the reference entity.

B) exposed to the credit risk of the protection seller.

C) said to be long the reference entity's credit risk.

Explanation

The credit protection buyer is exposed to the credit risk of the CDS seller. (Note that a CDS does not entirely eliminate credit risk; it eliminates the credit risk of the reference entity but substitutes it with the credit risk of the CDS seller.) The protection buyer is said to be *short* the reference entity's credit risk and is *bearish* on the financial condition of the reference entity.

(Module 29.1, LOS 29.a)

Question #10 of 20

Question ID: 1473605

A credit default swap (CDS) will change in value:

A) only when default occurs.

B) whenever the credit quality of the reference entity changes.

C) only when a failure to pay, a bankruptcy, or a restructuring occurs.

Explanation

CDS change in value over their lives as the credit quality of the reference entity changes; this leads to gains and losses for the CDS counterparties. This change in value will happen even though default may not have occurred – and even if it may never occur.

(Module 29.2, LOS 29.c)

Question #11 of 20

Question ID: 1473604

Credit default swap (CDS) fixed payments are *most* likely to:

A) be made until the maturity of the CDS whether a credit event occurs or not.

B) be made by the protection seller to the protection buyer .

C) be set at 1% for investment-grade debt and 5% for high-yield debt.

Explanation

CDS fixed payments are customarily set at a fixed annual rate of 1% for investment-grade debt or 5% for high-yield debt. Fixed payments are made by the CDS buyer to the CDS seller. The protection buyer is obligated to make regular payments until maturity of the CDS or until default (whichever occurs first).

(Module 29.2, LOS 29.c)

Question #12 of 20

Question ID: 1473588

Gill Westmore is the fixed income portfolio manager for Allied Insurance. Westmore has bought protection using a 2-year CDS on CDX-IG (125 constituent) index. The notional is $200 million. Company X, an index constituent defaults and trades at 25% of par.

The payoff on the CDS on account of default of X and the notional principal of the CDS after default are *closest* to:

	Payoff	Notional
A)	$1.5 million	$198 million
B)	$1.6 million	$200 million
C)	$1.2 million	$198.4 million

Explanation

Notional principal attributable to bonds of company X = $200 million/125 = $1.6 million.

Payoff on the CDS = $1.6 million − (0.25)($1.6 million) = $1.2 million.

After default, the CDS continues with (200-1.6) $198.4 million of notional principal.

(Module 29.1, LOS 29.a)

Question #13 of 20

Question ID: 1473589

Which of the following statements about credit default swaps (CDS) is *least* accurate? A credit default swap's reference obligation is:

A) the only obligation of the reference entity covered by a single-name CDS.

B) delivered by the protection buyer to the protection seller, upon default, in the case of physical settlement.

C) typically a senior unsecured bond.

Explanation

The reference obligation is not the only instrument covered by the CDS: any debt obligation issued by the borrower that is ranked equivalently ("pari passu") in priority of claims, or higher, relative to the reference obligation, is covered. A CDS's reference obligation is typically a senior unsecured bond. In the case of physical settlement, the reference obligation is delivered by the protection buyer to the protection seller, in exchange for the CDS notional.

(Module 29.1, LOS 29.a)

Question #14 of 20

Question ID: 1473601

Which of the following statements regarding settlement protocols with respect to CDS is *least* accurate?

A) When there is a credit event, the swap will be settled in cash or by physical delivery.

B) A super majority vote of the declarations committee of ISDA is needed for a credit event to be declared.

C) When a credit event has occurred, with physical settlement, the protection seller receives the reference obligation and the protection buyer receives the market value of the reference obligation immediately prior to the credit event.

Explanation

In case of physical settlement, the protection buyer receives the notional principal and not the market value of the bond prior to the credit event.

(Module 29.1, LOS 29.b)

Peter Nathan an asset manager for a hedge fund and looking to include credit default swaps (CDS) in the portfolio.

Nathan wants to know more about credit default swaps (CDS). He read a report that explained the characteristics of these products and the pricing theory. The report contained the following:

Comment 1: In a CDS, the protection buyer is long the credit risk of the reference entity.

Comment 2: In an index CDS, the lower the credit correlation, the cheaper the premium.

Nathan owns some intermediate-term bonds issued by ABC Company and has become concerned about the risk of a near-term default, although he is not very concerned about a default in the long term. ABC Company's two-year duration CDS currently trades at 400 bps, and the five-year duration CDS is at 700 bps.

Nathan evaluates the bonds of VAX and believes that some trading opportunities exist. The VAX bonds are currently trading at 260 bps above MRR in an asset swap, while the CDS premium is 200 bps.

Question #15 - 18 of 20

Question ID: 1473611

Which of the following *best* describe Comments 1 and 2?

A) Both comments are correct.

B) Both comments are incorrect.

C) Only one of the two comments is correct.

Explanation

Comment 1 is incorrect. It should say: "In a CDS, the protection buyer is short the credit risk of the reference entity." Note the CDS purchaser is typically referred to as the short party. Long and short for a CDS is relative to credit risk rather than buying or selling the instrument.

Comment 2 is correct. An Index CDS provides default cover for an index (basket) of bonds. Higher default correlation will make it more expensive to buy cover.

(Module 29.1, LOS 29.a)

Question #16 - 18 of 20

Question ID: 1473612

Assume that Nathan sells $400 million of protection on the equally weighted CDX IG index which consists of 125 entities. Concerned about the creditworthiness of an entity A, he purchases $2 million of single-name CDS protection on entity A. What is the investor's net notional exposure to Company A?

A) $2.0 million.

B) $1.2 million.

C) $3.2 million.

Explanation

The investor is long $3.2 million notional ($400 million / 125) through the index CDS and is short $2 million notional through the single-name CDS. His net notional exposure is $1.2 million.

(Module 29.1, LOS 29.a)

Question #17 - 18 of 20

Question ID: 1473613

Describe a potential curve trade that Nathan could use to hedge the default risk of ABC Company.

A) Nathan should position himself short in the short term CDS and short in the long term CDS.

B) Nathan should position himself long in the short term CDS and short in the long term CDS.

C) Nathan should position himself short in the short term CDS and long in the long term CDS.

Explanation

The investor anticipates a flattening curve and can exploit this possibility by positioning himself short (buying protection) in the two year CDS while going long in the five-year CDS (selling protection).

(Module 29.1, LOS 29.a)

Question #18 - 18 of 20

Question ID: 1473614

The *most* appropriate trade for the VAX bond is:

A) long the VAX bonds and buy the CDS.

B) short the VAX bonds and buy the CDS.

C) long the VAX bonds and sell the CDS.

Explanation

This is a basis trade where the strategy is to exploit price differences between the bond market and the CDS market. The transaction is an arbitrage based on credit risk priced into the two products.

Nathan will pick up 60 bps in yield when it buys the bond and buys the CDS. Nathan is also fully protected against credit risk.

(Module 29.3, LOS 29.e)

Question #19 of 20

Question ID: 1473608

Which of the following strategies would be *most appropriate* use of CDS given an expectation of credit curve steepening?

A) A curve flattening trade. ✗

B) Engage in a naked CDS. ✗

C) A curve steepening trade. ✓

Explanation

A credit curve steepening expectation would entail the credit spread for longer maturities increasing relative to the change in credit spread for shorter maturities. In such a scenario, one would buy protection for longer maturities and sell protection for shorter maturity (i.e., a curve steepening trade).

(Module 29.3, LOS 29.d)

Question #20 of 20

Question ID: 1473603

5-year, 5% Zillon Corp. bonds currently trade at $980 reflecting credit spread of 3%. A 5-year CDS for Zillon bonds has a coupon rate of 5%. The duration of the CDS = 4.

The upfront payment made/received by the protection buyer on a $4 million notional CDS is *closest* to:

A) $320,000 received by the protection buyer. ✓

B) $300,000 paid by the protection buyer. ✗

C) $400,000 received by the protection buyer. ✗

Explanation

Upfront payment	= (CDS spread − CDS coupon) × duration × notional principal
	= (0.03 − 0.05) × 4 × 4,000,000 = −$320,000

The protection buyer will receive an upfront premium of $320,000.

(Module 29.2, LOS 29.c)